

# Giovanna Kobus Conrado

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## RESEARCH INTERESTS

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Parameterized Algorithms, Graph Theory, Static Analysis, Automata Theory, Weak Memory Models, CFL Reachability

## EMPLOYMENT

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**Postdoctoral researcher**      Oct 2025 - Present  
*Aarhus University, Denmark*  
Supervised by Andreas Pavlogiannis

## EDUCATION

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**Doctor of Philosophy, Computer Science and Engineering**      Sep 2021-Sep 2025  
*Hong Kong University of Science and Technology, Hong Kong*  
Advisors: Amir Goharshady and Pedro Sander  
Thesis: “Parameterized Graph Algorithms and Their Role in Static Program Analysis”

**Bachelor’s Degree, Computer Science**      Feb 2017-Jul 2021  
*University of São Paulo, Brazil*  
Bachelor’s Thesis “Algorithms for the 2D ham sandwich problem”

## PUBLICATIONS & PREPRINTS

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In my field conferences are the typical venue. All published papers listed below are full peer-reviewed research papers. I use Australian CORE rankings when appropriate. Author order has not been modified.

**On the Decidability of Verification under Release/Acquire**      Preprint  
*G.K. Conrado, A. Pavlogiannis*

**Combinatorial Parameterized Algorithms for Chemical Descriptors based on Molecular Graph Sparsity**      LAGOS 2025  
*G.K. Conrado, A.K. Goharshady, H.J. Motwani, S. Novozhilov*

**CFL-based Methods for Approximating Interleaved Dyck Reachability**      STTT 2025  
*G.K. Conrado, A. Pavlogiannis*

**Program Analysis via Multiple Context Free Language Reachability**      POPL 2025

*G.K. Conrado, A. H. Kjelstrøm, A. Pavlogiannis, J. van de Pol*  
CORE A\*

**PYRAMID: A Protocol for Private and Trustless Multi-level  
Marketing on the Blockchain**

IEEE BCCA 2024

*G.K. Conrado, A.K. Goharshady, K.N. Long Nguyen*

**A Better Approximation for Interleaved Dyck Reachability**

SOAP 2024

*G.K. Conrado, A. Pavlogiannis*  
Best Presentation Award

**Faster Treewidth-based Approximations for Wiener Index**

SEA 2024

*G.K. Conrado, A.K. Goharshady, P. Hudec, Pingjiang Li, H.J. Motwani*  
CORE B

**The Bounded Pathwidth of Control-flow Graphs**

OOPSLA 2023

*G.K. Conrado, A.K. Goharshady, C.K. Lam*  
CORE A

**Exploiting the Sparseness of Control-flow and Call Graphs for  
Efficient and On-demand Algebraic Program Analysis**

OOPSLA 2023

*G.K. Conrado, A.K. Goharshady, K. Kochekov, Y.C. Tsai, A.K. Zaher*  
CORE A

## GRANTS

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**MSCA Postdoctoral Fellowships - Seal of Excellence**

2025

The project "Algorithmic Metatheory of Memory Models" was recognized as a high-quality project proposal following evaluation by an international panel of independent experts. This seal is awarded to projects that scored over 85% in the project evaluation but could not be funded due to budgetary constraints.

**Humboldt Research Fellowship for Postdocs (Declined)**

2025

Postdoctoral fellowship offered by the Alexander von Humboldt-Stiftung Foundation. The Fellowship provides an annual stipend EUR36,000 (approximately US\$42,500). It is a highly prestigious fellowship with acceptance rate between 20% and 25%.

**Hong Kong PhD Fellowship Scheme**

2021 - 2024

Fellowship established by the Research Grants Council of Hong Kong. The Fellowship provides an annual stipend of HK\$322,800 (approximately US\$41,400). It is highly competitive with a less than 2% acceptance rate.

**Redbird Scholarship Program**

2021

HKUST award in recognition of outstanding academic performance and research capacity.

## VISITS

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### **IIT Bombay**

January 2025-February 2025

#### *Research Intern*

Internship under the supervision of Prof. S. Akshay. Worked on the application of parameterization techniques in Program Synthesis and Petri Nets.

### **Aarhus University**

July 2023-January 2024

#### *Research Intern*

Internship under the supervision of Prof. Andreas Pavlogiannis. Worked mostly on problems related to interleaved Dyck reachability.

## TEACHING AND SUPERVISION

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### **TA, Computational Geometry**

Spring 2024

*Hong Kong University of Science and Technology, Hong Kong*

Prof. David Mount

### **Instructor, Saudi Arabia IOI camp**

February 2023

*Malik Abdulaziz Foundation for Giftedness and Creativity, Saudi Arabia*

### **TA, Advanced Algorithms**

Fall 2022

*Hong Kong University of Science and Technology, Hong Kong*

Prof. Amir Goharshady

### **TA, Design and Analysis of Algorithms**

Spring 2022

*Hong Kong University of Science and Technology, Hong Kong*

Prof. Dimitris Papadias

### **Instructor, ICPC Summer School**

January 2022

*Unicamp, Brazil*

### **Lecturer, Advanced Programming Challenges**

Spring 2021

*University of São Paulo, Brazil*

### **TA, Problem Solving Programming Strategies**

Spring 2019

*Texas A&M University, USA*

Prof. John Keyser

### **TA, Introduction to Computer Science**

Fall 2018

*University of São Paulo, Brazil*

Prof. Yoshiharu Kohayakawa

### **Instructor, UFRGS Winter School**

July 2018

*Universidade Federal do Rio Grande do Sul, Brazil*

### **TA, Mathematics for Computer Science**

Spring 2018

*University of São Paulo, Brazil*

Prof. Yoshiharu Kohayakawa

## SCHOOLS AND WORKSHOPS ATTENDED

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### **Czech Summer School in Discrete Mathematics**

July 2024

*Charles University, Prague, Czech Republic*

Attended courses “Combinatorial and algorithmic applications of twin-width” by Édouard Bonnet and “Poset inequalities” by Igor Pak

### **EPIT 2024 - Graphs and Algorithms: Conjectures**

May 2024

*CAES, Aussois, France*

Attended courses on width parameters, forbidden induced structures, coloring, distributed graph algorithms and games on graphs and hypergraphs

### **Swedish Summer School in Computer Science 2022**

June 2022

*KTH Stockholm, Sweden*

Completed courses “The Method of Moments in Computer Science and Beyond” by Ankur Moitra and “Polyhedral Techniques in Combinatorial Optimization” by Ola Svensson

### **XVII Summer School in Discrete Mathematics**

Jan 2022

*CMM - Center for Mathematical Modeling — FCFM — Universidad de Chile, Chile*

Completed courses “Graph Turán problems” by Boris Bukh, “Prophets, Secretaries, and other online puzzles” by Shuchi Chawla and “Submodular functions in combinatorial optimization” by Jan Vondrak

### **Paulista Workshop in Optimization, Combinatorics and Algorithms**

Nov 2020

Worked on the complexity of searching in random partial orders in the random graph model (problem proposed by Yoshikaru Kohayakawa)